ECON726: Policy and Program Evaluation

Tara Sullivan

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Outline

Causal Inference on Observational Data

Description: In real-world settings, analysts and policymakers must draw inferences about causes and effects from observational (non-randomized) data. For example:

- Does congestion pricing improve traffic in the central business district?
- Does zoning reform lead to more houses being built?
- Do speeding cameras make street safer for pedestrians?
- How do we truly determine whether charter schools produce better student outcomes?
- Whether a marketing campaign has increased consumer awareness?

This course will cover statistical methods for **causal inference** to help analysts design and analyze observational studies, for real-world decision-making.

- In plain language, causal inference involves using statistical tools and data to evaluate causal questions

Course goals

- Understand the central tenants of causal inference
- Challenge yourself to learn at least one causal methodology well (both theoretically and in practice)
  - The empirical project is something you should be able to talk about in job interviews
- Learn what it means to share runnable code
- Communicate results effectively, both in writing and as a presentation
- Learn some basic python (or: learn how to engage with code in a language you don't 100% understand)

Prerequisites

- This is a course in econometrics and applied statistics; as such, it is expected that you have a strong interest in business and public policy applications
- You are required to have taken a semester-long course in probability and statistical inference
  - Ideally, you have completed coursework in regression methods; otherwise, the econometric theory and the technical approach for the course will be difficult to follow

Prerequisites

- This is a course in econometrics and applied statistics; as such, it is expected that you have a strong interest in business and public policy applications
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- Statistical analysis software is the primary tool for this course, and I will be providing course instructions in **python**.

Grading

- Empirical project: 80% of final grade
- Homework/quizzes: 10% of final grade
- Attendance and participation: 10% of final grade

Empirical project

The project is based on an empirical paper of your choice which:

1. Uses a method discussed in this course*
2. Has data available online
3. Is not based on an experiment*

Your project should:

1. Be based on a paper published in a peer-reviewed journal
2. Briefly state the research question and discuss the dataset
3. Provide a (formal) discussion of the identification strategy (What is the ideal randomized experiment? What sort of bias would naïve approaches introduce? Why is this methodology causal?)
4. Replicate (selected) main findings
5. Describe and implement some sort of extension to this paper. This could be new and interesting tests for key assumptions, robustness checks, additional results based on other methods or data, replicating results in a different programming language, or combination thereof

Paper selection: Due **September 28** [5% of final grade]

- Select your paper. Briefly describe the causal effect of interest and methodology. Discuss what you think you will do for your extension of this paper.

Project plan: Due **October 12** [15% of final grade]

- This paper should be your first effort at addressing points (1) and (2) above. Summarize the public policy or business context of the paper. Write down the specific question the paper is evaluating. Discuss the data your are using (both your replication data, and data for your extension). Verify that you can access the data and replication package. Discuss the extensions you are considering, and, if necessary, where you will be getting that data. Verify that you've read the paper.

Replication results: Due **November 2**: [10% of final grade]

- You should submit some of your main replication results. The final results should be formatted as tables and charts in a word document or PDF, submitted on Brightspace. Your code and data should accessible on GitHub.

Presentation: Due **December 1** & **December 8** [10% of final grade]

- Prepare a 10 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 11** [40% of final grade]

- Your paper should contain 6-10 pages of text, and 3-6 pages of figures (double spaced, 12 point font). You must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Selecting a paper

Top 5 economics journals:

- American Economic Review
  - Note: American Economic Review: Papers and Proceedings are short papers and are NOT sufficient for this assignment
- Econometrica
- Journal of Political Economy
- Quarterly Journal of Economics
- Review of Economic Studies

Also good:

- American Economic Journal: Applied Economics
- Journal of Public Economics
- Review of Economics and Statistics

Data used in the Mostly Harmless Econometrics book:
`https://economics.mit.edu/people/faculty/josh-angrist/mhe-data-archive`

- Note that Krueger (1999) is based on an experiment, and would not be appropriate for this assignment

Good resource for finding papers (we'll discuss next class):
`https://paulgp.com/econlit-pipeline/index.html`

AI Policy

- This course is mean to extend your theoretical and empirical abilities. Over-reliance on LLMs will hamper that effort. Overall, my aim is to reward genuine effort, over seemingly perfect code and text.
- To that end, I highly encourage you to avoid AI for your work. I understand it may be helpful for brainstorming and troubleshooting, but please ensure that your written code and words are your own.
- Further, **unthinking AI usage will be heavily penalized**. If I can tell that you thoughtlessly used an AI, that will result in a zero on your assignment (or that part of your assignment).
- To be transparent about my AI usage: I will primarily use LLMs to brainstorm and troubleshoot. I've been advised by other professors to run your code through AI for grading purposes (as a secondary grader); if you would prefer I don't do that, please don't hesitate to let me know.
- Overall, this policy is meant to be a dialogue, not dogma. Please feel free to let me know your thoughts.

Course outline

Outline

1. Introduction/idealized experiment
2. Probability review
3. Statistics review
4. Potential outcomes and experiments
5. Linear regression models
6. Matching and Propensity models
7. Difference-in-differences models
8. Synthetic Control
9. Machine learning methods in Causal inference

Currently not listed: instrumental variables

- Currently not discussing instrumental variables (Angrist and Pischke 2009, Chapter 4)
- This can be a topic you explore in your empirical project

Resources

The primary textbook for this course is **Mostly Harmless Econometrics** (Angrist and Pischke, 2009). This textbook is meant to provide a readable introduction to many topics of this course. It is available though the Hunter College bookstore (link), though is cheaper on Amazon

A number of other resources exist on this topic:

Causal Inference for the Brave and True (Alves 2022)

- Excellent online companion to Mostly Harmless; I will cite from this website throughout the course. Effectively brings the methods from Mostly Harmless into the machine learning age, with sample code in Python. Be aware there are (very occasional!) errors.

Business Data Science (Taddy 2019)

- Sometimes called the "machine learning Mostly Harmless." I will borrow heavily from Chapters 5 (Experiments) and 6 (Controls). Note that the code provided in the book is written in R.

Applied Causal Inference Powered by ML and AI (Chernozhukov et al. 2025)

- Covers similar topics as Mostly Harmless, but from an ML perspective. Currently available free online.

Homework

Homework due next week (on Brightspace)

- Create a GitHub account (you can use a student account)

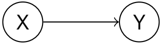
Things you should be working towards:

- Order Angrist and Pischke (2009); Chapter 1 & 2 (in 2 weeks)
- Consider which paper you would like to base your empirical project on (homework proposing paper ideas due in 2 weeks)

Schedule today

- High-level overview of potential outcomes framework, causal questions
- Brief review of methodologies I recommend you use for your paper (diff-in-diff and synthetic control)

Causal questions



Does X cause Y?

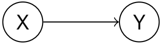
Causal questions

$$X \longrightarrow Y$$

Does X cause Y?

If X causes Y, how large is the effect of X on Y?

Causal questions



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Is the size of this effect large relative to the effects of other causes of Y?

Causal questions

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If individuals with $X = x'$ instead had $X = x''$, how much would their value of Y change?

Causal questions

$$X \longrightarrow Y$$

Does X cause Y?

If X causes Y, how large is the effect of X on Y?

Is the size of this effect large relative to the effects of other causes of Y?

If individuals with $X = x'$ instead had $X = x''$, how much would their value of Y change?

Examples of applied questions:

- What is the effect of class size on test scores?
- What is the effect of training programs on re-employment?
- How does schooling affect future earnings?

Examples of other causal questions?

Example: Effect of hospital visits on health

Suppose you want to know the effect of hospital visits on health. To that end, you look at some data on self-assessed health and hospital stays.

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Y	An individual's self-assessed health
$D \in \{0, 1\}$	1 if they stayed in the hospital in the past 12 months; 0 otherwise

Group	N (sample size)	$\bar{Y}$ (sample mean)	$\hat{\sigma}$ (standard error)
$D = 1$	7,774	3.21	0.0014
$D = 0$	90,049	3.93	0.003

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What if we compare the health of those admitted to the hospital to the health of those who weren't?

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$$\bar{Y}_1 - \bar{Y}_0 = 3.21 - 3.93 = -0.72$$

On average, the health of people who stayed in a hospital is **lower**

Do hospitals cause bad health?

Naïve comparison and selection problem

In general: Suppose you want to study the impact of a treatment on an outcome:

- Clinical example: the treatment could be a drug, and the outcome could be some health outcome
- Policy example: the treatment could be college attendance, and the outcome could be future wages

One solution: **naïve comparison**

- Look at people who receive the treatment, and people who do not, and measure differences in the outcome
- Average the wages of people who went to college, and the wages of people who didn't go to college, and take the difference

Problem: people who select to receive the treatment and those who select not to receive the treatment are often different

- When measuring causal effects, this leads to **selection bias**
- People who select to go to college may be more likely to have higher wages than those who do not, irrespective of college attendance.

Helpful framework for formalizing this problem and finding a solution: **potential outcomes framework**

Treatment assignment

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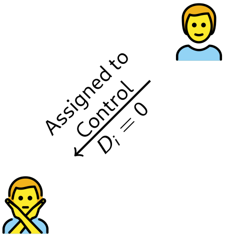

Treatment assignment

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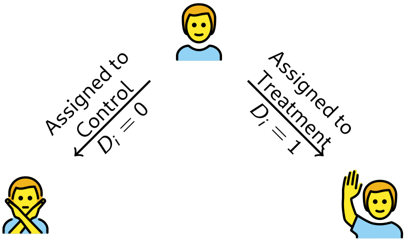
- Control group ($D_i = 0$)
- Treatment group ($D_i = 1$)



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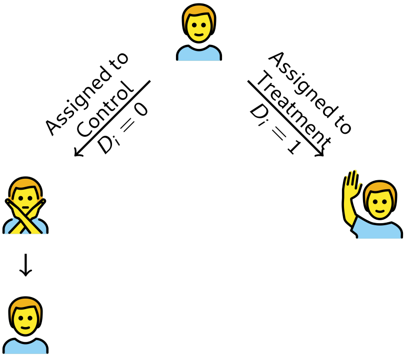

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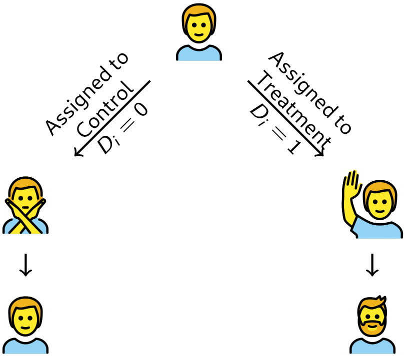

Potential outcome

Each participant i has a hypothetical outcome under treatment and control

- $Y_i(0)$: Outcome for individual i if they were assigned to the control group
- $Y_i(1)$: Outcome for individual i if they were assigned to the treated group

Participant	Control	Potential outcome	Treatment	Potential outcome
i	$D_i = 0$	$Y_i(0)$	$D_i = 1$	$Y_i(1)$
				

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Participant	Control	Potential outcome	Treatment	Potential outcome
i	$D_i = 0$	$Y_i(0)$	$D_i = 1$	$Y_i(1)$
👨			🙋‍♂️	🧔

Potential outcome

Each participant i has a hypothetical outcome under treatment and control

- $Y_i(0)$: Outcome for individual i if they were assigned to the control group
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Participant	Control	Potential outcome	Treatment	Potential outcome
i	$D_i = 0$	$Y_i(0)$	$D_i = 1$	$Y_i(1)$

Key: Each individual i in the data has a potential outcome under treatment and a potential outcome under control, irrespective of which group they are assigned to

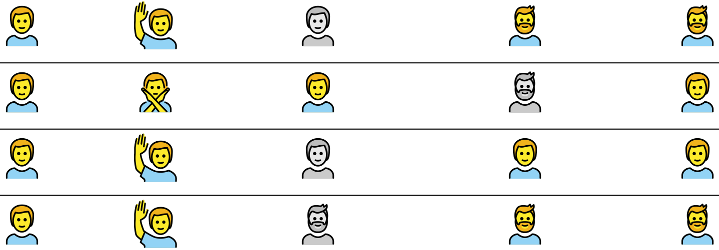
- For individuals assigned to the control group, their treatment potential outcome is unobservable (but hypothetically exists!)
- For individuals assigned to the treatment group, their control potential outcome is unobservable

Potential outcome and observable data

	Participant	Treatment D_i	Control potential outcome $Y_i(0)$	Treatment potential outcome $Y_i(1)$	Observed outcome Y_i
1	👨	🙋‍♂️	👨‍🦳	🧔	🧔
2	👨	🙅‍♂️	👨	🧔	👨
3	👨	🙋‍♂️	👨‍🦳	👨	👨
4	👨	🙋‍♂️	🧔	🧔	🧔

Causal effects

	Participant	Treatment D_i	Control potential outcome $Y_i(0)$	Treatment potential outcome $Y_i(1)$	Observed outcome Y_i
1					
2					
3					
4					



Causal effects of interest:

$$\text{Average treatment effect (ATE)} = \mathbf{E}[Y_i(1) - Y_i(0)]$$
$$\text{Average treatment effect on the treated (ATT)} = \mathbf{E}[Y_i(1) - Y_i(0) | D_i = 1]$$

Problem: You cannot observe potential outcomes $\{Y_i(0), Y_i(1)\}$ for everyone

- *Solution*: Randomization allows us to uncover certain causal effects

Potential outcomes under randomization

Randomization implies that the potential outcomes $\{Y_i(0), Y_i(1)\}$ are **independent** of treatment assignment D_i:

- Two variables are independent if the occurrence of one does not impact the occurrence of the other, and vice versa
- Nothing about your underlying potential outcomes determines likeliness to take the treatment

In this example, in a lab setting, independence of treatment assignment and potential outcomes means your assignment to treatment doesn't impact your control potential outcome or treatment potential outcome, and vice versa

- You have randomization of the treatment **assignment**;

You *would not* have independence of treatment and potential outcomes if participants got to choose whether they take the drug or not

- Only those who wanted to grow a beard would take the drug
- When participants can **select** whether they want to be treated, that leads to **selection bias** when estimating causal effects

Causal diagram

$$\begin{tikzcd} \boxed{\text{Treatment } (D_i)} \arrow[d] \\ \boxed{\text{Outcome } (Y_i)} \end{tikzcd}$$

Let's generalize our example a bit. Suppose we observe two variables:

- Some treatment dummy variable D_i, that takes on values 0 or 1
- Some outcome variable Y_i we're interested in

Goal: measure the effect that treatment ($D_i = 1$) has on outcome (Y_i).

- A similar lab experiment example, where D_i indicates whether someone takes a drug, and Y_i is some health outcomes

Assume:

- D_i is perfectly randomized (so D_i is independent of potential outcomes $\{Y_i(0), Y_i(1)\}$)
- The effect of D_i on Y_i is the same for the whole population
- No other variable influences Y_i (like in a perfect lab experiment)

Randomized experiment

Treatment (D_i)	**Participant**	**Outcome** (Y_i)
0	🙅‍♂️	Y_1
0	🙅‍♂️	Y_2
0	🙅‍♂️	Y_3
1	🙋‍♂️	Y_4
1	🙋‍♂️	Y_5
1	🙋‍♂️	Y_6

Treatment D_i is randomly assigned and **independent** of potential outcomes $\{Y_i(0), Y_i(1)\}$

- Occurrence of D_i does not impact potential outcomes, and vice versa

Can estimate the average treatment effect (ATE) using a **naïve comparison**:

Randomized experiment

Treatment (D_i)	**Participant**	**Outcome** (Y_i)
0	🙅‍♂️	Y_1
0	🙅‍♂️	Y_2
0	🙅‍♂️	Y_3
1	🙋‍♂️	Y_4
1	🙋‍♂️	Y_5
1	🙋‍♂️	Y_6

$$\mathbf{E}[Y|D=0] = \frac{1}{3}(Y_1 + Y_2 + Y_3)]$$

Treatment D_i is randomly assigned and **independent** of potential outcomes $\{Y_i(0), Y_i(1)\}$

- Occurrence of D_i does not impact potential outcomes, and vice versa

Can estimate the average treatment effect (ATE) using a **naïve comparison**:

- Average outcome for the control

Randomized experiment

Treatment (D_i)	**Participant**	**Outcome** (Y_i)
0	🙅	Y_1
0	🙅	Y_2
0	🙅	Y_3
1	🙋	Y_4
1	🙋	Y_5
1	🙋	Y_6

$$\mathbf{E}[Y|D=0] = \frac{1}{3}(Y_1 + Y_2 + Y_3)]$$

$$\mathbf{E}[Y|D=1] = \frac{1}{3}(Y_4 + Y_5 + Y_6)]$$

Treatment D_i is randomly assigned and **independent** of potential outcomes $\{Y_i(0), Y_i(1)\}$

- Occurrence of D_i does not impact potential outcomes, and vice versa

Can estimate the average treatment effect (ATE) using a **naïve comparison**:

- Average outcome for the control
- Average outcome for the treated

Randomized experiment

Treatment (D_i)	**Participant**	**Outcome** (Y_i)
0	🙅	Y_1
0	🙅	Y_2
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1	🙋	Y_4
1	🙋	Y_5
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$$\mathbf{E}[Y|D=0] = \frac{1}{3}(Y_1 + Y_2 + Y_3)]$$

$$\mathbf{E}[Y|D=1] = \frac{1}{3}(Y_4 + Y_5 + Y_6)]$$

$$\text{ATE} = \mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]$$

Treatment D_i is randomly assigned and **independent** of potential outcomes $\{Y_i(0), Y_i(1)\}$

- Occurrence of D_i does not impact potential outcomes, and vice versa

Can estimate the average treatment effect (ATE) using a **naïve comparison**:

- Average outcome for the control
- Average outcome for the treated
- Taking the difference

Observed differences in outcomes under potential outcome framework

Why do we need independence of treatment assignment (D_i) and potential outcomes ($\{Y_i(0), Y_i(1)\}$)?

- What we can see in the data is the **Observed difference** between groups

$$\underbrace{\mathbf{E}[Y_i|D_i = 1] - \mathbf{E}[Y_i|D_i = 0]}_{\text{Observed difference in outcomes}}$$

- What we want to know is the unobservable difference in potential outcomes:

$$\text{Average treatment effect (ATE)} = \mathbf{E}[Y_i(1) - Y_i(0)]$$
$$\text{Average treatment effect on the treated (ATT)} = \mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1]$$

To see why we need independence of treatment assignment and potential outcomes, it helps to use the **switching equation**:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i)Y_i(0)$$

$\Longrightarrow$ Switching equation lets you go between potential outcomes ($\{Y_i(0), Y_i(1)\}$) and observed outcome Y_i

Naïve comparison: $\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]$

Recall that:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i)Y_i(0)$$

Let's consider the averages of Y_{1i}, Y_{0i}, and Y_i:

$$\begin{aligned}
\mathbf{E}[Y_i|D_i=1] =& \mathbf{E}[D_i Y_i(1) + (1-D_i)Y_i(0)|D=1] \qquad \because Y_i = D_i Y_i(1) + (1-D_i)Y_i(0) \\
=& \mathbf{E}[Y_i(1)|D_i=1] \\
\mathbf{E}[Y|D=0] =& \mathbf{E}[Y_i(0)|D_i=0]
\end{aligned}$$

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Recall that:

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$$\begin{aligned}
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=& \mathbf{E}[Y_i(1)|D_i = 1] \\
\mathbf{E}[Y|D = 0] =& \mathbf{E}[Y_i(0)|D_i = 0]
\end{aligned}$$

We can re-write the observed difference as:

Naïve comparison: $\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]$

Recall that:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i)Y_i(0)$$

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=& \mathbf{E}[Y_i(1)|D_i=1] \\
\mathbf{E}[Y|D=0] =& \mathbf{E}[Y_i(0)|D_i=0]
\end{aligned}$$

We can re-write the observed difference as:

$$\underbrace{\mathbf{E}[Y_i|D_i=1] - \mathbf{E}[Y_i|D_i=0]}_{\text{Observed difference}} = \mathbf{E}[Y_i(1)|D_i=1] - \mathbf{E}[Y_i(0)|D_i=0]$$

Naïve comparison: $\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]$

Recall that:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i)Y_i(0)$$

Let's consider the averages of Y_{1i}, Y_{0i}, and Y_i:

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\mathbf{E}[Y_i|D_i = 1] =& \mathbf{E}[D_i Y_i(1) + (1 - D_i)Y_i(0)|D = 1] \qquad \because Y_i = D_i Y_i(1) + (1 - D_i)Y_i(0) \\
=& \mathbf{E}[Y_i(1)|D_i = 1] \\
\mathbf{E}[Y|D = 0] =& \mathbf{E}[Y_i(0)|D_i = 0]
\end{aligned}$$

We can re-write the observed difference as:

$$\begin{aligned}
\underbrace{\mathbf{E}[Y_i|D_i = 1] - \mathbf{E}[Y_i|D_i = 0]}_{\text{Observed difference}} =& \mathbf{E}[Y_i(1)|D_i = 1] - \mathbf{E}[Y_i(0)|D_i = 0] \\
=& \mathbf{E}[Y_i(1)|D_i = 1] - \mathbf{E}[Y_i(0)|D_i = 1] + \mathbf{E}[Y_i(0)|D_i = 1] \\
& - \mathbf{E}[Y_i(0)|D_i = 0]
\end{aligned}$$

Naïve comparison: $\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]$

Recall that:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i)Y_i(0)$$

Let's consider the averages of Y_{1i}, Y_{0i}, and Y_i:

$$\begin{aligned}
\mathbf{E}[Y_i|D_i=1] =& \mathbf{E}[D_i Y_i(1) + (1-D_i)Y_i(0)|D=1] \qquad \because Y_i = D_i Y_i(1) + (1-D_i)Y_i(0) \\
=& \mathbf{E}[Y_i(1)|D_i=1] \\
\mathbf{E}[Y|D=0] =& \mathbf{E}[Y_i(0)|D_i=0]
\end{aligned}$$

We can re-write the observed difference as:

$$\begin{aligned}
\underbrace{\mathbf{E}[Y_i|D_i=1] - \mathbf{E}[Y_i|D_i=0]}_{\text{Observed difference}} =& \mathbf{E}[Y_i(1)|D_i=1] - \mathbf{E}[Y_i(0)|D_i=0] \\
=& \mathbf{E}[Y_i(1)|D_i=1] - \mathbf{E}[Y_i(0)|D_i=1] + \mathbf{E}[Y_i(0)|D_i=1] \\
& - \mathbf{E}[Y_i(0)|D_i=0] \\
=& \mathbf{E}[Y_i(1) - Y_i(0)|D_i=1] + \mathbf{E}[Y_i(0)|D_i=1] - \mathbf{E}[Y_i(0)|D_i=0]
\end{aligned}$$

Naïve comparison: $\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]$

Recall that:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i)Y_i(0)$$

Let's consider the averages of Y_{1i}, Y_{0i}, and Y_i:

$$\begin{aligned}
\mathbf{E}[Y_i|D_i = 1] =& \mathbf{E}[D_i Y_i(1) + (1 - D_i)Y_i(0)|D = 1] \quad \because Y_i = D_i Y_i(1) + (1 - D_i)Y_i(0) \\
=& \mathbf{E}[Y_i(1)|D_i = 1] \\
\mathbf{E}[Y|D = 0] =& \mathbf{E}[Y_i(0)|D_i = 0]
\end{aligned}$$

We can re-write the observed difference as:

$$\begin{aligned}
\underbrace{\mathbf{E}[Y_i|D_i = 1] - \mathbf{E}[Y_i|D_i = 0]}_{\text{Observed difference}} =& \mathbf{E}[Y_i(1)|D_i = 1] - \mathbf{E}[Y_i(0)|D_i = 0] \\
=& \mathbf{E}[Y_i(1)|D_i = 1] - \mathbf{E}[Y_i(0)|D_i = 1] + \mathbf{E}[Y_i(0)|D_i = 1] \\
& - \mathbf{E}[Y_i(0)|D_i = 0] \\
=& \underbrace{\mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1]}_{\text{Avg. treatment effect on treated}} + \underbrace{\mathbf{E}[Y_i(0)|D_i = 1] - \mathbf{E}[Y_i(0)|D_i = 0]}_{\text{Selection bias}}
\end{aligned}$$

Causal effects in randomized experiment

Observed difference in outcomes equals causal effect of interest plus selection bias:

$$\underbrace{\mathbf{E}[Y_i|D_i=1]-\mathbf{E}[Y_i|D_i=0]}_{\text{Observed difference in outcomes}}=\underbrace{\mathbf{E}[Y_i(1)-Y_i(0)|D_i=1]}_{\text{Avg. treatment effect on treated}}+\underbrace{\mathbf{E}[Y_i(0)|D_i=1]-\mathbf{E}[Y_i(0)|D_i=0]}_{\text{Selection bias}}$$

Consider the section bias when measuring the effect of college attendance on wages:

- $\mathbf{E}[Y_i(0)|D_i=1]$: the wage someone went to college would have *if they hadn't gone*
- $\mathbf{E}[Y_i(0)|D_i=0]$: the wage of someone who didn't go to college

These objects are equal under independence of treatment assignment (D_i) and potential outcomes ($\{Y_i(0), Y_i(1)\}$)

Further, under randomization ATT = ATE! (this is less important than the selection point above, but a nice-to-have)

$$\begin{aligned}
\mathbf{E}[Y_i|D_i=1]-\mathbf{E}[Y_i|D_i=0] =& \mathbf{E}[Y_i(1)|D_i=1]-\mathbf{E}[Y_i(0)|D_i=0]\\
=& \mathbf{E}[Y_i(1)|D_i=1]-\mathbf{E}[Y_i(0)|D_i=1]\\
=& \mathbf{E}[Y_i(1)-Y_i(0)|D_i=1]\\
=& \mathbf{E}[Y_i(1)-Y_i(0)]\\
=& \mathbf{E}[Y_i(1)]-\mathbf{E}[Y_i(0)]
\end{aligned}$$

Selection bias

$$\underbrace{\mathbf{E}[Y_i|D_i=1]-\mathbf{E}[Y_i|D_i=0]}_{\text{Observed difference in outcomes}}=\underbrace{\mathbf{E}[Y_i(1)-Y_i(0)|D_i=1]}_{\text{Avg. treatment effect on treated}}+\underbrace{\mathbf{E}[Y_i(0)|D_i=1]-\mathbf{E}[Y_i(0)|D_i=0]}_{\text{Selection bias}}$$

BUT there's often reason to believe that selection bias in non-zero

- What if people who go to college are "higher ability" (broadly defined)

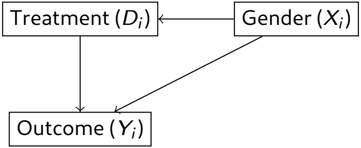
$\implies$ They would earn a higher wage than those in the control group, even if they hadn't gone to college:

$$\mathbf{E}[Y_i(0)|D_i=1]>\mathbf{E}[Y_i(0)|D_i=0]$$

$\implies$ Selection bias is positive

Let's consider a simplified case where selection bias is non-zero

Causal effects under randomization with covariates



Now suppose:

- D_i: dummy variable (like taking a drug in an experiment)
- Y_i: outcome (like whether the drug causes patients to grow facial hair)
- X_i: whether the participant is female ($X_i = 1$ means female)
  - To keep things simple, we assume X_i is a single binary variable, but it could represent many covariates, discrete and continuous

Assume:

- D_i is perfectly randomized **by gender**
  - This means potential outcomes $\{Y_i(0), Y_i(1)\}$ are independent of treatment D_i, **conditional on** X_i
- The effect of D_i on Y_i is the same for everyone within a gender
- No other variable influences Y_i

Causal effects with data

$$
\begin{tikzcd}
\boxed{\text{Treatment } (D_i)} \arrow[d] & \boxed{\text{Gender } (X_i)} \arrow[l] \arrow[dl] \\
\boxed{\text{Outcome } (Y_i)} &
\end{tikzcd}
$$

Then we can estimate the average treatment effect (ATE) by gender from the observed outcomes:

- $\mathbf{E}[Y_i|D_i = 1, X_i = 1]$: Average of observed outcomes Y when the participant is female and takes the drug
- $\mathbf{E}[Y_i|D_i = 0, X_i = 1]$: Average of observed outcomes Y when the participant is female and does not take the drug

ATE for females is difference between these two values:

$$\text{ATE} = \mathbf{E}[Y_i|D_i = 1, X_i = 1] - \mathbf{E}[Y_i|D_i = 0, X_i = 1]$$

Randomization by covariate X

Treatment (D_i)	**Participant**	**Female** (X_i)	**Outcome** (Y_i)
0	🙅‍♂️	0	Y_1
0	🙅‍♂️	0	Y_2
1	🙋‍♂️	0	Y_3
1	🙋‍♂️	0	Y_4
0	🙅‍♀️	1	Y_5
0	🙅‍♀️	1	Y_6
1	🙋‍♀️	1	Y_7
1	🙋‍♀️	1	Y_8

Randomization by covariate X

Treatment (D_i)	**Participant**	**Female** (X_i)	**Outcome** (Y_i)
0	🙅‍♂️	0	Y_1
0	🙅‍♂️	0	Y_2
1	🙋‍♂️	0	Y_3
1	🙋‍♂️	0	Y_4
0	🙅‍♀️	1	Y_5
0	🙅‍♀️	1	Y_6
1	🙋‍♀️	1	Y_7
1	🙋‍♀️	1	Y_8

Average effect for females:

$$\delta_{X_i=1} = \mathbf{E}[Y_i|D_i = 1, X_i = 1] - \mathbf{E}[Y_i|D_i = 0, X_i = 1] = \frac{1}{2}(Y_7 + Y_8) - \frac{1}{2}(Y_5 + Y_6)$$

Randomization by covariate X

Treatment (D_i)	**Participant**	**Female** (X_i)	**Outcome** (Y_i)
0	🙅‍♂️	0	Y_1
0	🙅‍♂️	0	Y_2
1	🙋‍♂️	0	Y_3
1	🙋‍♂️	0	Y_4
0	🙅‍♀️	1	Y_5
0	🙅‍♀️	1	Y_6
1	🙋‍♀️	1	Y_7
1	🙋‍♀️	1	Y_8

Average effect for females:

$$\delta_{X_i=1} = \mathbf{E}[Y_i|D_i=1, X_i=1] - \mathbf{E}[Y_i|D_i=0, X_i=1] = \frac{1}{2}(Y_7 + Y_8) - \frac{1}{2}(Y_5 + Y_6)$$

Average effect for males:

$$\delta_{X_i=0} = \mathbf{E}[Y_i|D_i=1, X_i=0] - \mathbf{E}[Y_i|D_i=0, X_i=0] = \frac{1}{2}(Y_3 + Y_4) - \frac{1}{2}(Y_1 + Y_2)$$

Randomization by covariate X

Treatment (D_i)	**Participant**	**Female** (X_i)	**Outcome** (Y_i)
0	🙅‍♂️	0	Y_1
0	🙅‍♂️	0	Y_2
1	🙋‍♂️	0	Y_3
1	🙋‍♂️	0	Y_4
0	🙅‍♀️	1	Y_5
0	🙅‍♀️	1	Y_6
1	🙋‍♀️	1	Y_7
1	🙋‍♀️	1	Y_8

Population average effects can be found by weighing the effects for males ($\delta_{X_i=0}$) and for females ($\delta_{X_i=1}$):

- This is a weighted average, with weights determined by the number of males and females in the sample

Randomization by covariate X

Treatment (D_i)	**Participant**	**Female** (X_i)	**Outcome** (Y_i)
0	🙅‍♂️	0	Y_1
0	🙅‍♂️	0	Y_2
1	🙋‍♂️	0	Y_3
1	🙋‍♂️	0	Y_4
0	🙅‍♀️	1	Y_5
0	🙅‍♀️	1	Y_6
1	🙋‍♀️	1	Y_7
1	🙋‍♀️	1	Y_8

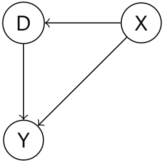
Ability to estimate effects hinges on **independence** of treatment and potential outcome **conditional on** X

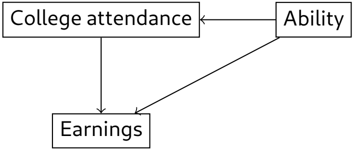
- Given gender (X_i), treatment (D_i) is randomly assigned

$\Longrightarrow$ In policy questions, we want to find covariates X_i, such that treatment LOOKS randomly assigned; this is the **conditional independence assumption (CIA)**

Conditional independence assumption

Conditional independence assumption: occurrence of treatment D_i does not influence the occurrence of potential outcomes $\{Y_{0i}, Y_{1i}\}$, or vice versa, conditional on knowing X_i:





Consider measuring the impact of college attendance on future earnings:

- Suppose people are "high ability" or "low ability"
- Assume treatment (college attendance) is as good as randomly assigned, conditional on ability, X_i

$\Longrightarrow$ You can estimate the effect of college attendance on wages by:

- Finding the average effect for high ability people
- Finding the average effect for low ability people
- Weighing these average effects, based on the distribution of ability in the population

Selection bias under conditional independence assumption (CIA)

Observed difference in outcomes equals causal effect of interest plus selection bias:

$$\underbrace{\mathbf{E}[Y_i|D_i=1,X_i]-\mathbf{E}[Y_i|D_i=0,X_i]}_{\text{Observed difference in outcomes given } X_i}$$

$$=\underbrace{\mathbf{E}[Y_i(1)-Y_i(0)|D_i=1,X_i]}_{\text{Avg. treatment effect on treated}}+\underbrace{\mathbf{E}[Y_i(0)|D_i=1,X_i]-\mathbf{E}[Y_i(0)|D_i=0,X_i]}_{\text{Selection bias}}$$

Under CIA, selection bias disappears conditional on X_i:

$$\underbrace{\underbrace{\mathbf{E}[Y_i(0)|X_i,D_i=1]}_{\text{Control potential outcome for treated group}} = \underbrace{\mathbf{E}[Y_i(0)|X_i,D_i=0]}_{\text{Outcome for control group}}}_{\text{Selection bias equals zero}}$$

$\implies$ Difference in observed outcomes **conditional on** X_i equals causal effect of interest:

$$\mathbf{E}[Y_i|X_i,D_i=1]-\mathbf{E}[Y_i|X_i,D_i=0]=\mathbf{E}[Y_i(1)-Y_i(0)|D_i=1,X_i]$$

Given the CIA, comparisons of observed outcomes Y_i (conditional on X_i) have a causal interpretation

- Holding X_i contact, treatment assignment **looks** random
- Selection bias disappears

Non-randomized data

So far we have covered:

- Potential outcomes framework, and the fundamental problem of causal inference
- Measuring causal effects under randomization
- Measuring causal effects under the conditional independence assumption
  - Conditional on covariates X_i, treatment **looks** like its randomized

In policy, we can rarely estimate clean treatment effects from naïve comparisons:

1. Randomization is difficult, if not impossible
2. Other variables X (including unobservable variables) may impact treatment D
3. Measuring Y, X, and even D might be imperfect

Policy analysis

Key for policy analysis: **conditional independence assumption**

- Consider what the ideal experiment is, even if you can't estimate
- Determine how you can find other variables X_i such that treatment *looks* random (i.e. apply the conditional independence assumption)
- Estimate treatment effects

Methods for getting around non-randomized data?

- Differences-in-differences
- Synthetic control

Differences-in-differences

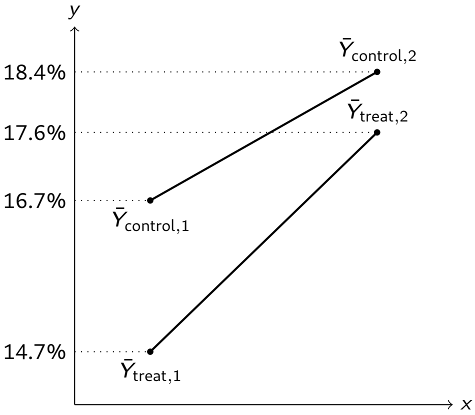
Suppose we have a workforce training program that is meant to increase employment.

- $D \in \{0, 1\}$: Treatment; $D = 1$ means i participates in the program
- Y: Whether or not i is employed

We can average the employment rate before and after the program for the treatment and control group:

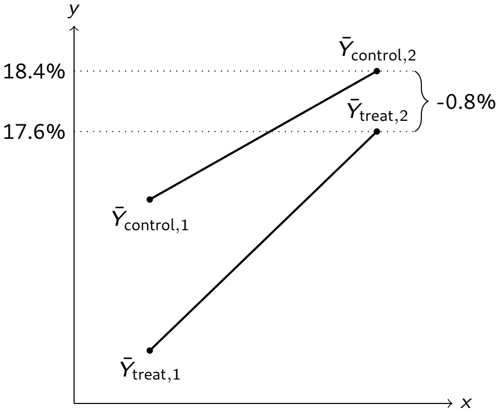
	$t = 1$	$t = 2$
Control group ($D = 0$)	14.7%	17.6%
Treatment group ($D = 1$)	16.7%	18.4%

Graphical representation of differences-in-differences



Question: How do we evaluate this treatment?

Differences in outcomes

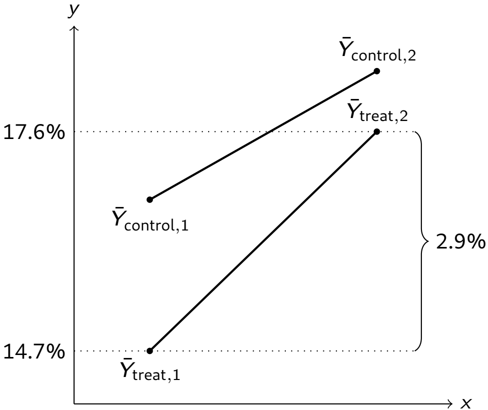


Method 1: Look at the employment rate for participants after the program

$$\begin{aligned}\bar{Y}_{\text{treat},2} - \bar{Y}_{\text{control},2} \\ &= 17.6 - 18.4 \\ &= -0.8\%\end{aligned}$$

- It seems like this training program caused employment to go down
- Pretty unsatisfying answer, participants in the treatment group started a lower employment rate than in the control group

Differences in time

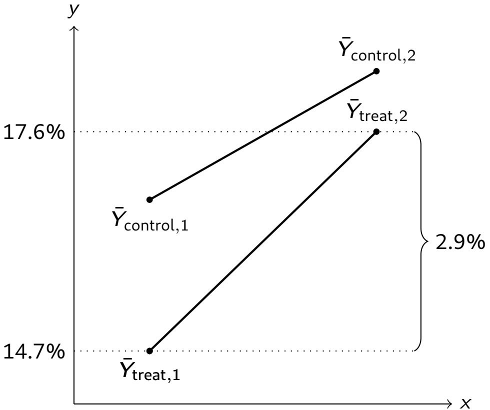


Method 2: Pre-post evaluation

$$\begin{aligned}\bar{Y}_{\text{treat},2} - \bar{Y}_{\text{treat},1} &= 17.6 - 14.7 \\ &= 2.9\%\end{aligned}$$

- It seems like this training program caused employment to rise
- Also an unsatisfying answer; employment in the control group also rose over time
- Maybe some external force acting on everyone (both control and treatment) that leads to higher employment rates
- Some of the improvement in the treatment group may be due to these external factors, not just the treatment

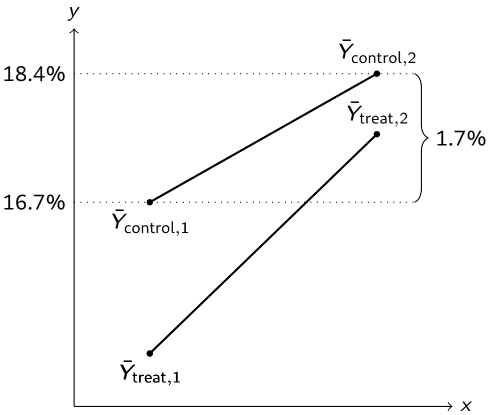
Differences in differences



Method 3: Compute the difference in employment for the treatment group before and after the program:

$$\begin{aligned}\bar{Y}_{\text{treat},2} - \bar{Y}_{\text{treat},1} &= 17.6 - 14.7 \\ &= 2.9\%\end{aligned}$$

Differences in differences



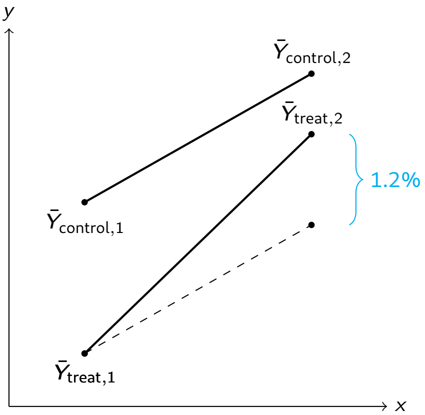
Method 3: Compute the difference in employment for the treatment group before and after the program:

$$\begin{aligned}\bar{Y}_{\text{treat},2} - \bar{Y}_{\text{treat},1} &= 17.6 - 14.7 \\ &= 2.9\%\end{aligned}$$

Compute the difference for the control group before and after the program:

$$\begin{aligned}\bar{Y}_{\text{control},2} - \bar{Y}_{\text{control},1} &= 18.4 - 16.7 \\ &= 1.7\%\end{aligned}$$

Differences in differences



Method 3: Compute the difference in employment for the treatment group before and after the program:

$$\begin{aligned}\bar{Y}_{\text{treat},2} - \bar{Y}_{\text{treat},1} =& 17.6 - 14.7\\ =& 2.9\%\end{aligned}$$

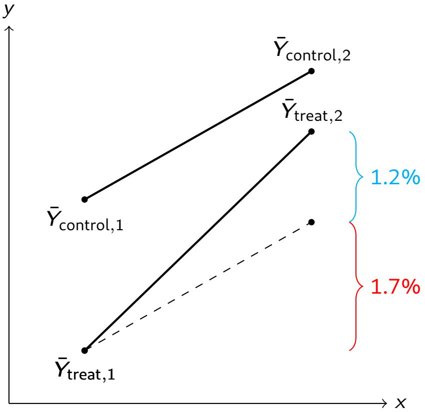
Compute the difference for the control group before and after the program:

$$\begin{aligned}\bar{Y}_{\text{control},2} - \bar{Y}_{\text{control},1} =& 18.4 - 16.7\\ =& 1.7\%\end{aligned}$$

Compute the differences between these differences:

$$2.9\% - 1.7\% = 1.2\%$$

Differences in differences



2.9%: Total increase in treatment group
1.7%: Increase that would have happened anyway
1.2%: Treatment effect

Method 3: Compute the difference in employment for the treatment group before and after the program:

$$\bar{Y}_{treat,2} - \bar{Y}_{treat,1} = 17.6 - 14.7 = 2.9\%$$

Compute the difference for the control group before and after the program:

$$\bar{Y}_{control,2} - \bar{Y}_{control,1} = 18.4 - 16.7 = 1.7\%$$

Compute the differences between these differences:

$$2.9\% - 1.7\% = 1.2\%$$

Synthetic Control

Abadie, Diamond, and Hainmueller: Synthetic Control Methods for Comparative Case Studies

Concern about health risks lead to the passage of Proposition 99 in California in 1988

- First modern tobacco control program in the United States
- Created cigarette excise tax of 25 cents per pack, and earmarked the tax revene for health and anti-smoking causes
- Following its passage, laws were enacted prohibiting smoking in restaurants, workplaces
- Largely viewed as having successfully cut smoking in California

Question: What was the effect of Prop 99 on cigarette consumption in California?

- **Problem:** California received the treatment. What control can you compare California to?

Finding a control

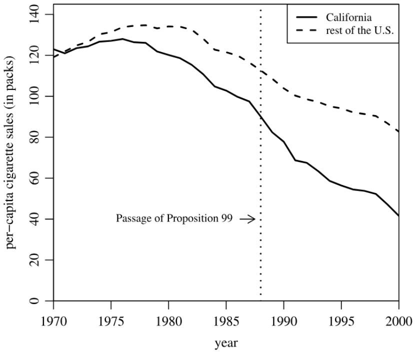


Figure 1. Trends in per-capita cigarette sales: California vs. the rest of the United States.

The rest of the United States may not provide a suitable comparison group for California

- Before the ban, cigarette consumption in California differed from the rest of the US

Building a synthetic California

Create a donor pool of states from which to build a synthetic California. Include all states that:

- Did not adopt large-scale tobacco control programs during the sample period
- Did not raise cigarette taxes by 50 cents or more in the post period

Construct a synthetic California as a weighted average of potential control states

- Weights should be chosen so that the resulting synthetic California best reproduces the values of a set of predictor of cigarette consumption in California before the passage of Prop 99

State weights in synthetic California

Table 2. State weights in the synthetic California

State	Weight	State	Weight
Alabama	0	Montana	0.199
Alaska	–	Nebraska	0
Arizona	–	Nevada	0.234
Arkansas	0	New Hampshire	0
Colorado	0.164	New Jersey	–
Connecticut	0.069	New Mexico	0
Delaware	0	New York	–
District of Columbia	–	North Carolina	0
Florida	–	North Dakota	0
Georgia	0	Ohio	0
Hawaii	–	Oklahoma	0
Idaho	0	Oregon	–
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.334
Maryland	–	Vermont	0
Massachusetts	–	Virginia	0
Michigan	–	Washington	–
Minnesota	0	West Virginia	0
Mississippi	0	Wisconsin	0
Missouri	0	Wyoming	0

Cigarette sales predictor means

Table 1. Cigarette sales predictor means

Variables	California Real	California Synthetic	Average of 38 control states
Ln(GDP per capita)	10.08	9.86	9.86
Percent aged 15–24	17.40	17.40	17.29
Retail price	89.42	89.41	87.27
Beer consumption per capita	24.28	24.20	23.75
Cigarette sales per capita 1988	90.10	91.62	114.20
Cigarette sales per capita 1980	120.20	120.43	136.58
Cigarette sales per capita 1975	127.10	126.99	132.81

NOTE: All variables except lagged cigarette sales are averaged for the 1980–1988 period (beer consumption is averaged 1984–1988). GDP per capita is measured in 1997 dollars, retail prices are measured in cents, beer consumption is measured in gallons, and cigarette sales are measured in packs.

Synthetic California

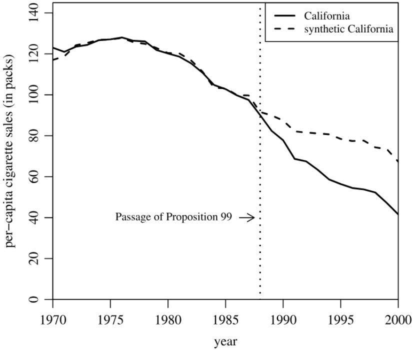


Figure 2. Trends in per-capita cigarette sales: California vs. synthetic California.